Project Report

DATA ANALYTICS WITH POWER BI

(Project Semester: August–December 2025)

CAR ACCIDENT ANALYSIS DASHBOARD

Submitted by

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DECLARATION

I, **Ritik Raushan**, student of B.tech under CSE/IT discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 20/12/2025
Signature: Ritik Raushan
Registration No: 12300915

CERTIFICATE

This is to certify that **Ritik Raushan**, bearing Registration No. **12300915**, has completed the **Data Science Minor Project** titled **"Car Accident Analysis Dashboard"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort, and study.

Signature of Supervisor
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Date: 20/12/2025

ACKNOWLEDGEMENT

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1. INTRODUCTION

Road traffic accidents are one of the leading causes of injuries, fatalities, and economic losses across the world. With rapid urbanization, increasing vehicle ownership, and complex traffic environments, the frequency and severity of road crashes have risen significantly. Understanding the underlying factors behind these accidents—such as driver behavior, vehicle characteristics, road conditions, and environmental factors—is essential for improving road safety and reducing risks.

This project, **"Car Accident Analysis Dashboard"** focuses on analyzing crash reporting data using **Power BI** to extract meaningful insights from large-scale accident records. The study aims to transform raw crash data into interactive visual dashboards that help identify accident trends, driver fault patterns, injury severity, and high-risk vehicles. By leveraging data analytics and visualization techniques, the project supports data-driven decision-making for road safety planning, awareness, and preventive measures.

2. SCOPE OF THE ANALYSIS

The scope of this project covers a comprehensive analysis of crash-related data to understand patterns, risks, and contributing factors. The analysis includes:

- Studying crash trends over different time periods to identify seasonal or periodic variations
- Analyzing driver behavior, including driver fault and driving direction
- Identifying high-risk vehicle manufacturers and models based on crash involvement
- Examining injury severity levels and types of crashes
- Evaluating the impact of speed limits and traffic control measures on crash occurrence
- Designing interactive Power BI dashboards for easy exploration and insights
- Supporting road safety analysis and policy-level decision-making

The project is limited to exploratory and descriptive analytics using historical data and does not include real-time data ingestion.

3. EXISTING SYSTEM

In traditional systems, crash data analysis is primarily performed using static reports, spreadsheets, or manual statistical methods. These approaches rely heavily on tabular data and textual summaries, making it difficult to quickly identify trends, correlations, or risk factors. Such systems often require significant time and effort for data interpretation and lack advanced visualization support.

Drawbacks of the Existing System

- Static and non-interactive reports
- Manual and time-consuming data analysis
- Limited visualization and storytelling capabilities
- Difficulty in identifying hidden risk patterns
- No support for trend-based or predictive insights

These limitations highlight the need for an interactive and visual analytics solution.

4. SOURCE OF DATASET

The dataset used in this project is the **Crash Reporting – Drivers Dataset**, obtained from the official U.S. government open data portal:

https://catalog.data.gov/dataset/crash-reporting-drivers-data

This dataset contains detailed records related to road crashes, including:

- Crash date and time
- Driver demographics and involvement
- Injury severity and fatality information
- Vehicle make, model, year, and body type
- Collision types and crash categories
- Traffic control conditions
- Speed limits and road-related information

The dataset is reliable, structured, and suitable for large-scale analytical visualization.

5. DATA PREPROCESSING / CLEANING

Before performing analysis, the dataset underwent multiple preprocessing steps to ensure data quality and consistency. These steps included:

- Removing duplicate and irrelevant records
- Handling missing and null values using appropriate methods
- Standardizing categorical values (e.g., collision types, vehicle body types)
- Correcting incorrect data types (date, numeric, categorical)
- Creating calculated columns and DAX measures
- Selecting relevant features required for analysis

These preprocessing steps improved data accuracy and ensured reliable analytical results.

6. ANALYSIS ON DATASET

i. Introduction

Multiple analytical techniques were applied to understand crash patterns, driver behavior, and vehicle risk factors using Power BI.

ii. General Description

The analysis primarily focuses on crash frequency, injury severity distribution, driver fault contribution, vehicle involvement, and speed-related risk assessment.

iii. Specific Requirements, Functions & Formulas

The following Power BI DAX measures and calculations were used:

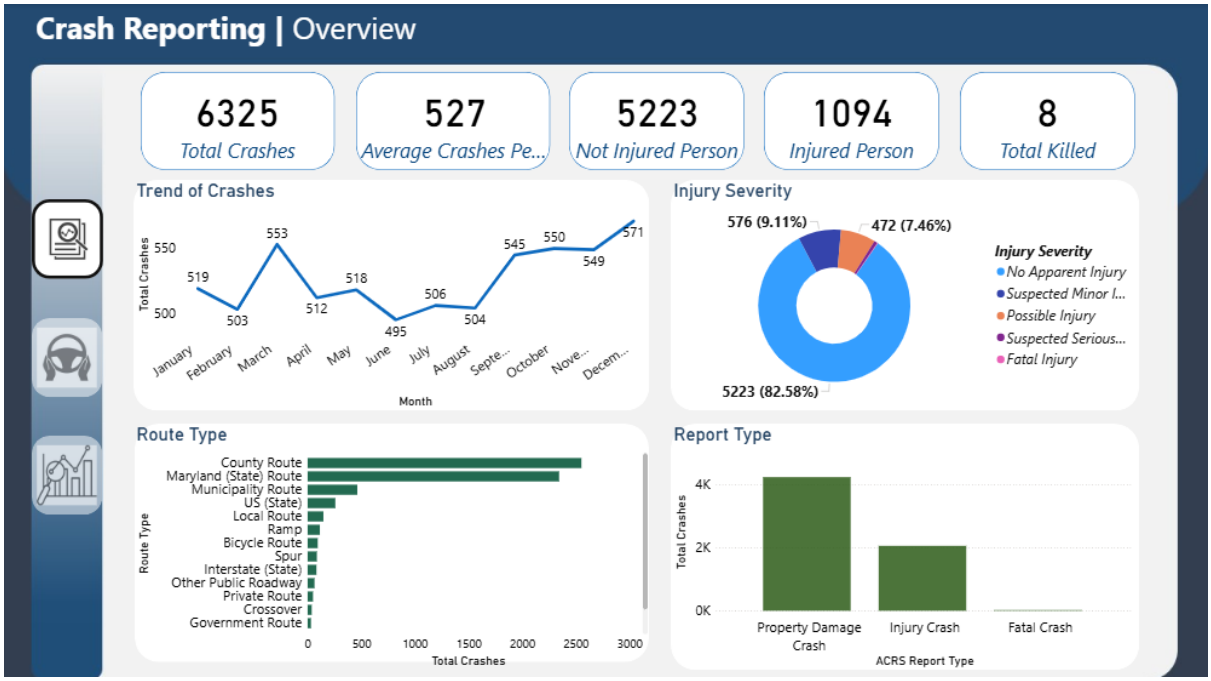
- Total Crashes
- Injury and Fatality Counts
- Average Crashes per Time Period
- Driver at Fault Percentage
- Top 10 High-Risk Vehicle Models and Manufacturers

iv. Analysis Results

- Passenger cars are involved in the highest number of crashes
- Front-to-rear collisions are the most common collision type
- The majority of crashes result in non-fatal injuries
- Certain vehicle models and manufacturers show higher crash involvement
- Mid-range speed limits show higher crash frequencies

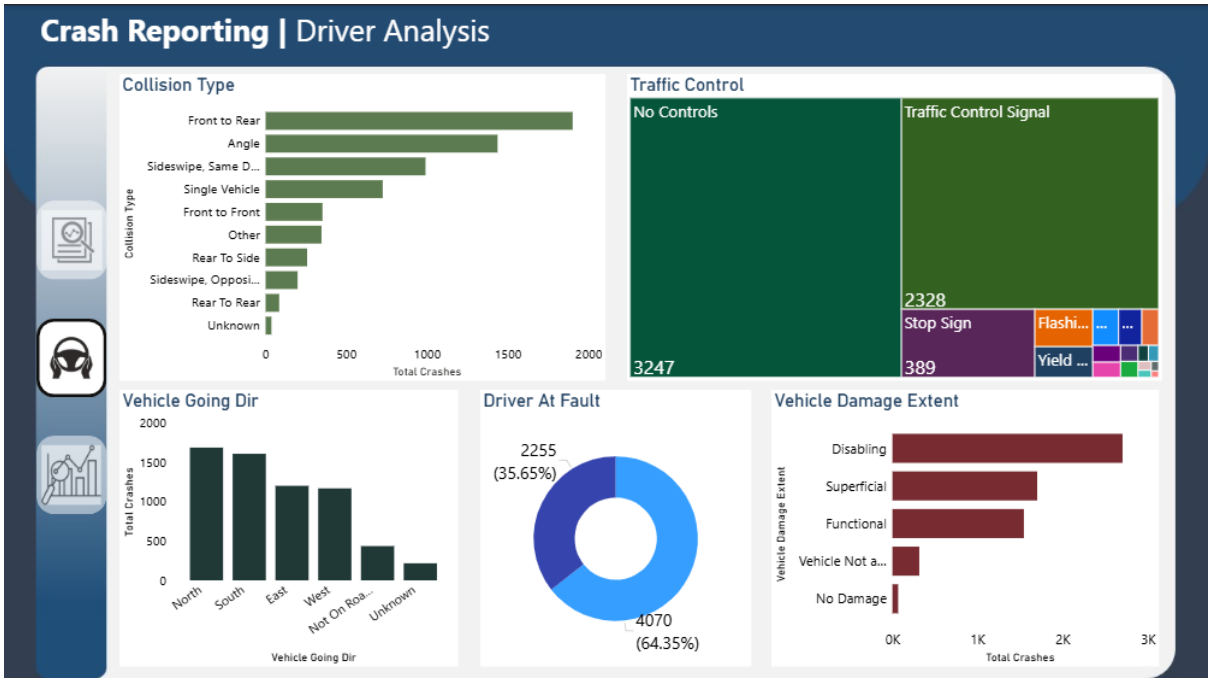
v. Dashboard Visualization

- **Overview Dashboard Page**



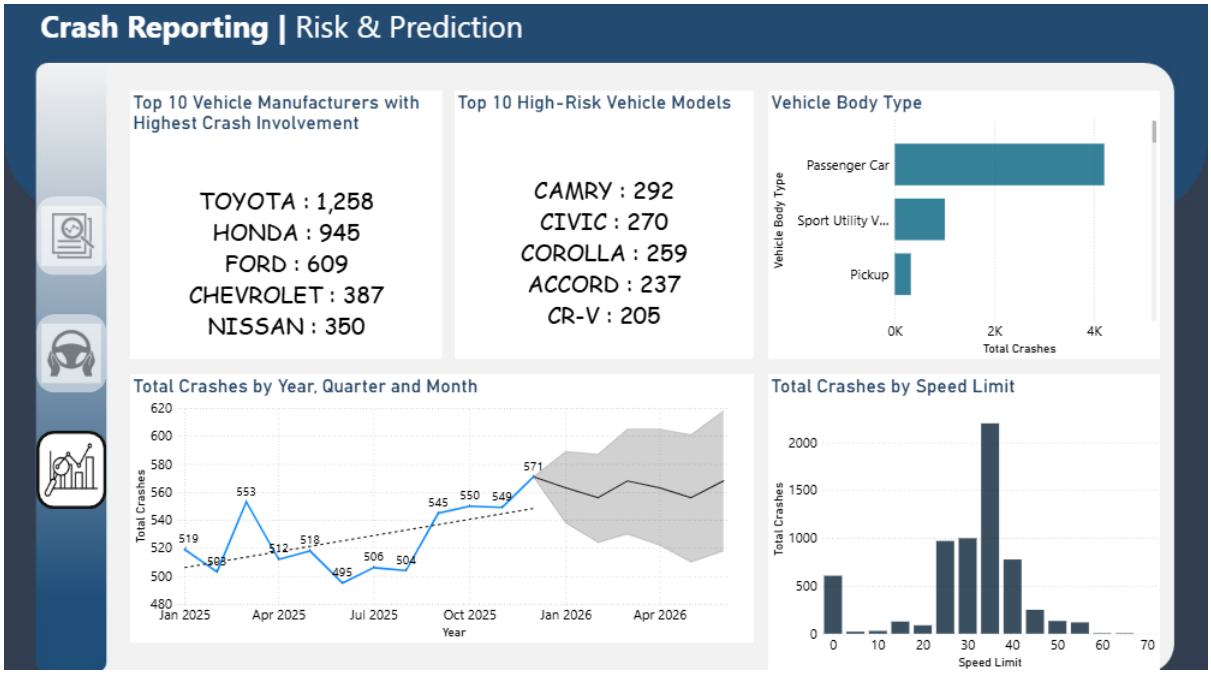
This dashboard provides a high-level summary of total crashes, injuries, fatalities, and crash trends over time using KPI cards and line charts.

- **Driver Analysis Dashboard Page**



This page analyzes collision types, driver fault percentage, vehicle direction, and traffic control conditions to understand driver behavior patterns.

- **Risk & Prediction Dashboard Page**



This dashboard identifies high-risk vehicle manufacturers and models, analyzes speed limits, and displays predictive crash trends.

7. LIST OF ANALYSIS WITH RESULTS

Analysis Type	Result
Crash Trend Analysis	Seasonal variations observed
Injury Severity Analysis	Majority of injuries are non-fatal
Driver Fault Analysis	Drivers responsible in most cases
Vehicle Risk Analysis	Certain models show higher crash risk
Speed Limit Analysis	Higher crashes at mid-range speeds

8. FUTURE SCOPE

The project can be further enhanced in the future by:

- Integrating real-time crash data sources
- Applying advanced machine learning models for crash prediction

- Performing accident hotspot analysis using GIS mapping
- Developing a driver risk scoring system
- Providing policy-level safety recommendations for authorities

9. REFERENCES

1. Microsoft Power BI Official Documentation
2. Government Crash Reporting Portals
3. Road Safety and Traffic Analysis Research Papers
4. IEEE Publications on Data Analytics

Dataset Link:

https://catalog.data.gov/dataset/crash-reporting-drivers-data

LinkedIn Link:

https://www.linkedin.com/posts/ritik314_powerbi-dataanalytics-crashanalysis-activity-7408110139045691392-ekqs?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEUcgtkBBVgcAr4YfW0zKas6vIkNdRFAFkA

GitHub Link:

https://github.com/RitikRaushan314/Car-Accident-Analysis-Dashboard/blob/main/Report.pdf